

VO3 Virtual Game Development Team

(formerly VO3 Software Engineering Team)

Description & Eligibility

Teams will design and develop an original endless-runner video game that performs and behaves as outlined in the provided topic. Project submissions will consist of game source code, art and audio assets, a Game Manual, and a functional, executable version of the game. This national event will be submitted and judged virtually. Awards will be presented at the National Leadership Conference. Any middle level, secondary or postsecondary contestant may enter this event. A team will consist of 2-4 contestants. There are no restrictions on the number of entries per chapter or per state. Teams who do not submit an entry following the topic listed below or do not follow the Copyright and/or Fair Use Guidelines will be disqualified.

Topic

Afterburn – Endless Runner. Create an endless runner game titled "Afterburn." Players control a young inventor riding a self-built rocket-powered skateboard through a sprawling futuristic city after an experimental energy surge sends the board into overdrive. As malfunctioning robots, collapsing scaffolding, and rogue delivery drones flood the streets, players must dash, jump, and slide through procedurally generated urban obstacles – collecting scattered power crystals to keep the board running and unlocking new gadgets along the way. Each run generates a unique city route, ensuring no two attempts are the same.

Competition Notes

- AI Policy Level 2 *Permitted with Citation*
- Contest Registration
 - Advisors register contestants for the event using the Membership Registration System, accessible at <https://register.bpa.org>
 - Click on the “Conferences” tab and then select the “2027 National Virtual Events”
 - There will be a \$20 fee for each team entry (*invoiced during event registration*)
 - Event registration can be purchased with your original invoice or with a new invoice after the team registration
 - The registration deadline is no later than 11:59 p.m. Eastern Time, on December 1, 2026
 - Contestants participating in the competition must be registered for the event prior to the submission deadline to be eligible
 - Only the team captain should complete the submission

Technical Pre-Submit Specifications

What to Submit at https://upload.bpa.org	Saved File Name(s)	National Deadline
Submit the following in one combined PDF: 1. Clickable URL to Project Files (game source code, art and audio assets, game manual, and functional game) 2. Works Cited 3. BPA Release Form(s) 4. AI Usage and Documentation Form	VO3-TeamID.pdf	January 15, 2027, 11:59 p.m. Eastern Time

Contest Competencies

1. Design and develop an original endless-runner video game that meets the requirements of the assigned topic, applying core game development principles across code, content, and design
2. Implement core game architecture, including the game loop, state management (menus, gameplay, pause, game over, restart), and persistent data for high scores, unlocked gadgets, and player settings

3. Apply game-specific mathematics and logic, including coordinate systems, vector movement, physics, collision detection, and basic AI behavior for obstacles and non-player characters
4. Design and implement procedural generation systems that produce varied, fair, and well-paced runs aligned with the topic's endless-runner format
5. Develop and tune core player mechanics (dash, jump, slide, boost, gadget use) so they feel responsive, learnable, and consistent with the topic
6. Create or curate cohesive game art, animation, and visual effects that establish a clear art direction and support gameplay readability
7. Design and implement audio for the game, including sound effects, music, and audio feedback that reinforce game feel and atmosphere
8. Design user interface and user experience elements (menus, HUD, onboarding) that clearly communicate vital information during fast-paced gameplay
9. Balance gameplay difficulty, pacing, and progression to deliver a fair, engaging, and re-playable experience
10. Apply object-oriented or component-based design, code documentation, and performance optimization techniques appropriate to game development
11. Evaluate project requirements and delegate responsibilities effectively within a team, managing scope, scheduling, and collaborative development across disciplines (code, art, audio, design)
12. Apply structured problem-solving and critical thinking to identify, analyze, and resolve technical and design challenges throughout development
13. Document the project clearly, including a Game Manual covering controls, mechanics, features, and build/launch instructions

Contest Specifications for this Event

- Original endless-runner gameplay built around the topic, featuring continuous forward motion and procedurally generated routes
- Core player actions include dash, jump, and slide, plus at least one gadget or boost mechanic earned through progression
- A menu interface for non-gameplay interaction (e.g., starting a run, resuming, exiting, viewing high scores, viewing/equipping unlocked gadgets)
- An on-screen HUD displaying vital gameplay information (e.g., distance, score, fuel/energy, active gadget)
- A minimum average framerate of 30 frames per second (FPS) during typical play
- Persistent storage of player progression data (high scores, unlocked gadgets, settings) saved and loaded between sessions
- Optional databases or persistence options include local file storage (JSON, XML, SQLite), Oracle Express, Microsoft SQL Server, Oracle MySQL, PostgreSQL, Google Firebase, or Amazon Web Services. Local persistence is acceptable
- Languages/frameworks permitted include Java, JavaScript, Python, Ruby, Microsoft C#, Objective-C/Cocoa, C++, Apple Swift, Unity, Unreal Engine, Godot, GameMaker, and other freely available game engines or libraries
- Any submission using a game development framework must clearly show the extent of original work done via source code comments
- Use of integrated development environments (e.g., Microsoft Visual Studio, Eclipse, Oracle NetBeans, or Apple XCode) is encouraged
- Submit the URL to all project documents including, but not limited to game source code, art and audio assets, a Game Manual, and a functional, executable version of the game. Works Cited, BPA Release Form(s), AI Usage and Documentation Form in a combined PDF file to <https://upload.bpa.org> no later than 11:59 p.m. Eastern Time, on January 15, 2027.
- Advisors of qualifying teams will be contacted to reserve a time for a videoconference presentation before a panel of judges. The dates and times for presentation reservations will be

announced via email notification. Teams will be assigned to sections prior to the preliminary presentation.

- Length of event: no more than three (3) minutes for setup, no more than ten (10) minutes for the presentation, and no more than five (5) minutes for judges' questions.
- The top contestants with the highest cumulative technical and presentation scores will be invited to attend the National Leadership Conference to be recognized for their outstanding efforts and to participate in the National Showcase.

Common Specifications for all BPA Judged Events

- Contestant ID is required for all submissions. Materials from non-registered contestants and/or those missing Contestant ID cannot be accepted.
- The team is responsible for securing a BPA Release Form(s) from any person, including themselves, whose image or work is used in the project.
- The team is responsible for citing all sources, including themselves, on the Works Cited page for any work used on the project. It is the policy of Business Professionals of America to comply with copyright law. The *Style & Reference Manual* contains guidelines for Copyright and/or Fair Use. Contestants will be disqualified for violations of the guidelines.
- Materials must follow the organization's BPA Graphic Standards and make proper use of the BPA logo and/or organization's name. Refer to the BPA Graphic Standards in the *Style & Reference Manual*.
- Judges' comments will be returned digitally through the online judging system at the national level. Materials submitted for technical judging will *not* be returned and will *not* be available at NLC.

V03 Virtual Game Development Team

Technical Scoring Rubric

Required Elements (If any question results in a NO, please assign a score of 0)				
Team followed the topic.				☐ Y ☐ N
Team followed the Copyright and/or Fair Use Guidelines and Artificial Intelligence Usage & Citation Policy.				☐ Y ☐ N
Team submission includes a game source code, art and audio assets, a Game Manual, and a functional, executable version of the game.				☐ Y ☐ N
If answered no to any question, please stop scoring and provide a brief reason for the <i>disqualification</i> :				
Submission and Formatting (All points or none are awarded by the technical judge)				
Team submitted one (1) combined PDF file containing the URL to the Project, Works Cited (formatted according to the <i>Style & Reference Manual</i>), BPA Release Form(s), and AI Usage and Documentation Form.				/10
Core Gameplay & Mechanics	Below Expectations 0-10 points	Meets Expectations 11-20 points	Above Expectations 21-30 points	Points Awarded
Core Run Mechanics (Dash, Jump, Slide)	Core actions (dash, jump, slide) are missing, broken, or feel unresponsive. Player movement does not match the endless-runner pacing of the topic.	Core actions are implemented and functional. Inputs are mostly responsive and produce expected results, though timing or feel may be inconsistent.	Core actions are tight, responsive, and well-tuned. Movement feels intentional and matches the high-speed endless-runner energy described in the topic.	/30
Procedural Level / Route Generation	City route is static, repetitive, or generation is broken. Runs feel identical and the topic requirement is not met.	Route is procedurally generated and produces meaningfully different runs. Variety is present but obstacle placement may feel uneven or repetitive over time.	Route generation produces unique, well-paced runs every time. Obstacle density, spacing, and difficulty curve feel deliberate and varied.	/30
Obstacle & Hazard Design	Obstacles (malfunctioning robots, scaffolding, rogue drones) are missing, indistinguishable, or unfair. Hits feel arbitrary.	Multiple obstacle types are present and behave as expected. Hit detection works in most cases, though some hazards may feel cheap or unclear.	A varied roster of obstacle types tied to the topic is present, each with clear visual telegraphing and fair, consistent collision behavior.	/30
Power Crystals & Gadget Unlocks	Crystal collection and/or gadget unlock systems are missing or non-functional. No meaningful progression exists.	Crystals can be collected and gadgets can be unlocked. Systems work but progression pacing or reward variety may feel limited.	Crystal collection, fuel/energy management, and gadget unlocks form a cohesive, satisfying progression loop directly aligned with the topic.	/30
Game Architecture & Code Quality	Below Expectations 0-8 points	Meets Expectations 9-17 points	Above Expectations 18-25 points	Points Awarded
Game Loop & State Management	Game loop is broken or unclear. Game states (menu, play, pause, game over) are missing, conflated, or behave unpredictably.	Game loop runs reliably and core states (menu, play, pause, game over) function correctly with minor inconsistencies.	Game loop is well-architected. State transitions (menu, play, pause, game over, restart) are clean, deterministic, and easy to follow in code.	/25

V03 Virtual Game Design Development Team

Presentation Scoring Rubric (Cont'd)

Object-Oriented / Modular Design	Code is monolithic or poorly structured. Classes, prefabs, or modules are missing, unclear, or have overlapping responsibilities.	Code is organized into classes/modules with mostly clear responsibilities. Some coupling or duplication may exist.	Code shows thoughtful OOP or component-based design with strong cohesion, clear responsibilities, and effective use of inheritance/composition where appropriate.	/25
Performance & Frame Rate	Game drops well below 30 FPS in normal play, stutters on routine actions, or fails to release memory (visible slowdowns over time).	Game maintains roughly 30 FPS in typical conditions. Occasional dips occur under load but do not break gameplay.	Game consistently meets or exceeds 30 FPS, including during heavy procedural generation. Demonstrates clear optimization (pooling, culling, asset management).	/25
Code Documentation & Readability	Comments are missing or misleading. Code is hard to follow and naming is inconsistent.	Most classes, methods, and complex logic are commented. Naming is generally consistent and code is mostly readable.	Code is well-documented with clear comment blocks on classes, key methods, and non-obvious logic. Naming, formatting, and structure consistently aid understanding.	/25
Art, Animation & Visual Design	Below Expectations 0-6 points	Meets Expectations 7-13 points	Above Expectations 14-20 points	Points Awarded
Art Direction & Visual Cohesion	Visual style is inconsistent or unfinished. Art assets clash and do not communicate the futuristic-city setting of the topic.	Visual style is generally consistent. Art assets support the futuristic-city setting, though some elements feel mismatched or placeholder.	Art direction is cohesive, intentional, and clearly communicates the futuristic-city setting. Color, scale, and style work together to support gameplay readability.	/20
Character & Object Animation	Animations are missing, stiff, or broken. The player character and obstacles do not visibly respond to input or state changes.	Core animations (run, jump, slide, dash) are present and functional. Transitions may be abrupt but communicate the action.	Animations are smooth, expressive, and clearly communicate every player action and obstacle behavior. Transitions feel polished and reinforce game feel.	/20
Visual Effects & Game Feel	Effects (particles, screen shake, trails, impact feedback) are missing or distract from gameplay.	Effects are present and support key moments (crystal collection, collisions, boosts). Some opportunities for feedback may be missed.	Effects enhance game feel meaningfully – collecting, crashing, and boosting all have satisfying, readable visual feedback that elevates the experience.	/20
Audio Design	Below Expectations 0-6 points	Meets Expectations 7-13 points	Above Expectations 14-20 points	Points Awarded
Sound Effects	Sound effects are missing, broken, or inappropriate. Key player actions and game events have no audio response.	Sound effects are present for major actions and events. Quality and timing are functional but may be inconsistent.	Sound effects are well-mixed, well-timed, and clearly reinforce player actions, hazards, pickups, and outcomes.	/20

Music & Audio Atmosphere	Music is missing or clashes with the setting. Audio atmosphere does not match the futuristic-city, high-speed feel of the topic.	Music is present and generally supports the setting. Loops or transitions may be noticeable or repetitive.	Music is original or well-licensed, fits the futuristic-city energy of the topic, and transitions smoothly across menu and gameplay states.	/20
UI, UX & Progression	Below Expectations 0-8 points	Meets Expectations 9-17 points	Above Expectations 18-25 points	Points Awarded
Menus & In-Game HUD	Menus are missing, broken, or confusing. HUD does not communicate vital information (score, distance, fuel/energy, gadgets).	Menus (start, pause, resume, exit) and HUD are present and functional. Some readability or hierarchy issues may exist.	Menus and HUD are clean, readable, and well-organized. Vital gameplay information is communicated clearly at a glance during fast-paced runs.	/25
Onboarding, Save Data & Progression Persistence	No tutorial or onboarding is present, and player progress, high scores, or unlocked gadgets are not saved between sessions.	Basic onboarding exists and core progression (high score, unlocked gadgets) is saved between sessions. Some elements may be missing or inconsistent.	Onboarding clearly teaches mechanics, and persistent data (high scores, unlocked gadgets, settings) reliably saves and loads across sessions.	/25
Stability & Technical Polish	Below Expectations 0-5 points	Meets Expectations 6-10 points	Above Expectations 11-15 points	Points Awarded
Stability & Error Handling	Game crashes during normal play, exceptions are unhandled, or errors disrupt the player experience.	Game runs reliably with occasional minor issues. Errors are caught in most expected scenarios but feedback to the player may be limited.	Game runs stably across a full session. Exceptions are caught gracefully, edge cases are handled, and any errors are communicated cleanly to the player.	/15
Balance & Difficulty Curve	Difficulty curve is broken – game is too easy, too hard, or spikes unfairly. Player has no chance to learn before failing.	Difficulty progresses logically over a run. Some pacing or spike issues may exist but the curve is generally fair.	Difficulty curve is well-tuned. Each run feels fair, learnable, and steadily more challenging, encouraging mastery and replay.	/15
TOTAL TECHNICAL POINTS				/400

VO3 Virtual Game Design Development Team

Presentation Scoring Rubric

Required Elements (If any question results in a NO, please assign a score of 0)				
Team followed the topic.				☐ Y ☐ N
Team followed Copyright and/or Fair Use Guidelines and Artificial Intelligence Usage & Citation Policy.				☐ Y ☐ N
If answered no to either question, please stop scoring and provide a brief reason for the <i>disqualification</i> :				
Items To Evaluate	Below Expectations 0-3 points	Meets Expectations 4-6 points	Above Expectations 7-10 points	Points Awarded
Design & Development Process	Process is unclear or incomplete; lacks structure or detail.	Process is explained with basic clarity and covers major phases.	Process is clearly explained with insight into design decisions and iteration.	/10
Game Design & Mechanics	Mechanics are poorly explained, disconnected, or do not match the topic.	Mechanics are described with basic clarity and relate to the topic.	Mechanics are clearly explained, well-integrated, and demonstrate deliberate design tied to the topic.	/10
Art, Animation & Visual Style	Art and visual decisions are mentioned but not explained or justified.	Art and visual decisions are described with some connection to gameplay and theme.	Art direction, animation, and visual style are clearly explained, justified, and tied to the game's identity.	/10
Audio & Sound Design	Audio choices are not discussed or are unjustified.	Audio choices are described and connect to gameplay or mood.	Audio choices are clearly explained and shown to reinforce game feel, atmosphere, and player feedback.	/10
Level / Procedural Generation Approach	Level or procedural generation approach is unclear or technically vague.	Approach is described with general accuracy; the team can speak to how runs vary.	Approach is clearly explained, technically sound, and demonstrates intentional design choices for variety, pacing, and fairness.	/10
Technical Architecture & Implementation	Architecture, languages, frameworks, and tools are not discussed clearly.	Team explains key technical choices (engine, language, structure) with basic accuracy.	Team clearly explains architecture, key systems, and trade-offs, demonstrating strong technical understanding of their build.	/10
Team Contribution to Development	Team roles are unclear or unevenly represented during the presentation.	Each team member's role and contribution is described with basic clarity.	Each team member clearly describes their contribution; the team demonstrates strong, balanced collaboration.	/10
Presentation Quality, Q&A and Style	Presentation lacks organization or clarity; responses to judges are vague or off-topic.	Presentation is organized and key points are communicated; responses to judges are mostly clear and relevant.	Presentation is polished, engaging, and professionally delivered; responses to judges are confident and show strong understanding.	/10

VO3 Virtual Game Design Development Team

Presentation Scoring Rubric (Cont'd)

Team Members	Set up lasted no longer than three (3) minutes and the presentation lasted no more than ten (10) minutes.	/10
Timing	At least two original members of the team in attendance at the time of the presentation.	/10
TOTAL PRESENTATION POINTS		/100
Scoring Category		Points Awarded
Technical Scoring Points		/400
Presentation Points		/100
TOTAL		/500

Props and/or additional items shall not be used as a basis for scoring.

PRESENTATION WILL BE STOPPED AT TEN (10) MINUTES